

SAR1130P Overview

80-43035-8 Rev. AA

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets

NO PUBLIC DISCLOSURE PERMITTED: Please report postings of this document on public servers or websites to: DocCtrlAgent@qualcomm.com.

© Qualcomm Technologies, Inc. and/or its subsidiaries. All rights reserved.

Revision History

Revision	Date	Description
AA	August 2025	Initial release

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Contents

1 SAR1130P Chipset	<u>4</u>
2 SAR1130P Reference Design Platforms	<u>10</u>
3 SAR1130P Hardware Features – CPU	<u>12</u>
4 SAR1130P Hardware Features – DSP	<u>14</u>
5 SAR1130P Hardware Features – Display	<u>16</u>
6 SAR1130P Hardware Features – GPU	<u>18</u>
7 SAR1130P Hardware Features – Audio	<u>20</u>
8 SAR1130P Hardware Features – Camera	<u>23</u>
9 SAR1130P Hardware Features – Video	<u>27</u>
10 SAR1130P Hardware Features – Security	<u>29</u>
11 SAR1130P Sensor Modules	<u>31</u>
12 References	<u>34</u>

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

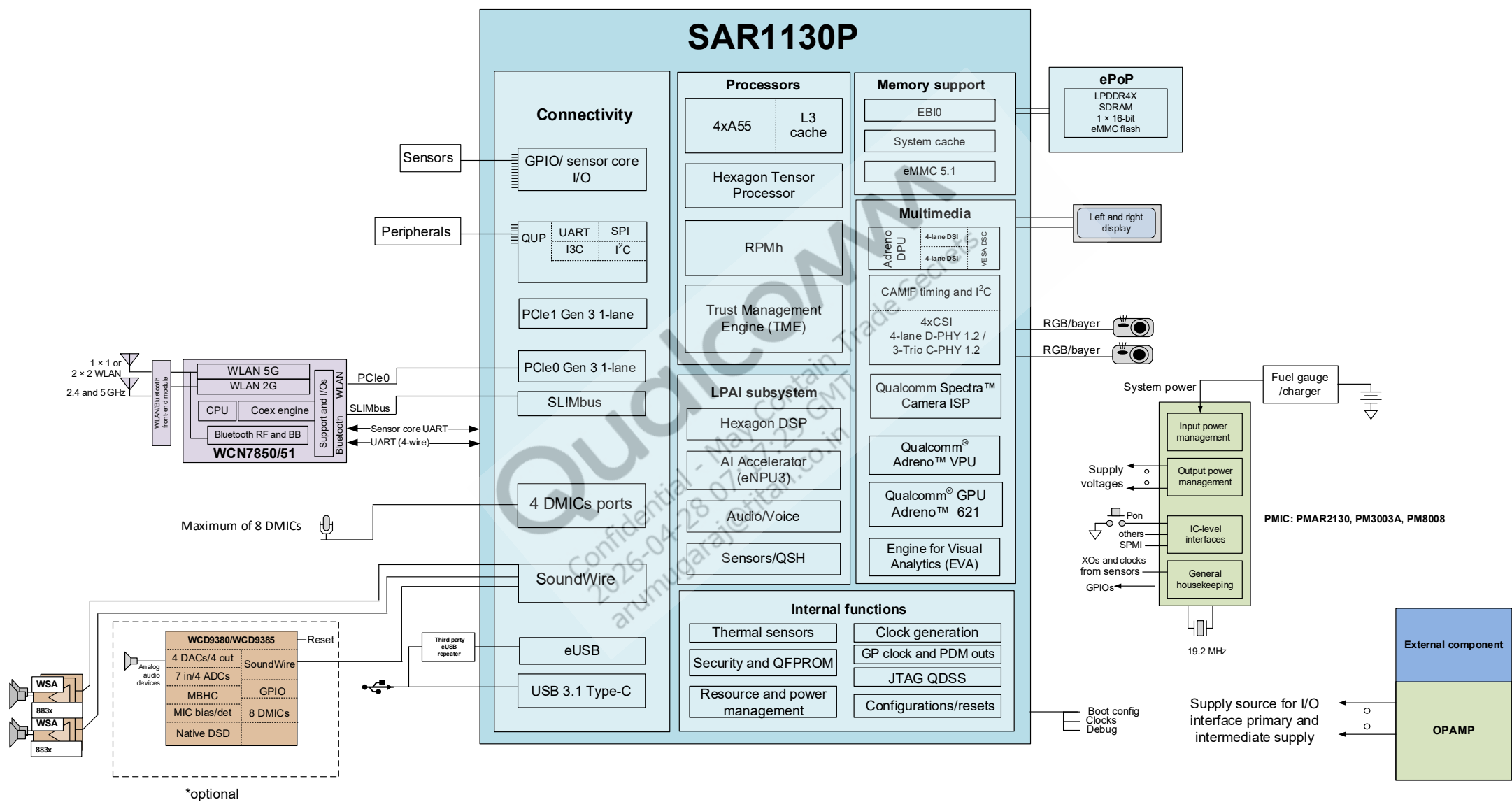


Section 1

SAR1130P Chipset

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

SAR1130P Block Diagram



SAR1130P Chipset IC Combination

- For information on schematics and pin assignment, see
 - SAR1130P Reference Schematic (DP25-43766-100A / DP25-43766-101)
 - SAR1130P Pin Assignment and GPIO Configuration Spreadsheet (80-43766-1A)
 - SAR1130P Digital Baseband Design and Layout Guidelines/Training Slides (80-43766-5)

SAR1130P: AP/Wi-Fi/audio subsystems

Processor	PMIC	Wi-Fi (11ax)/Bluetooth	Audio
SAR1130P	PMAR2130	WCN7851 Dual band radio, supporting 802.11a/b/g/n/ac/ax/be + BT 5.3	Codec – WCD9380/WCD9385 Speaker amplifier – WSA8830/WSA8835

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Hardware Feature Comparison vs. SXR1130

Features	SXR1130	SAR1130P
Process/package	10 nm/11.1 × 10.5 mm	4 nm/10.0 × 12.0 mm
CPU	8 × Qualcomm® Kryo™ CPU 300, L3 1 MB: 2 × Gold (2.5 GHz), 6 × Silver (1.7 GHz)	4 Kryo Silver, L3 1.5 MB
Memory	Dual-channel non-PoP high-speed memory, LPDDR4X SDRAM designed for 1866 MHz clock (2 × 16 bit)	1 x 16 single-channel embedded package-on-package (ePoP) high-speed LPDDR4X SDRAM 2133 MHz clock
Display	1080p at 30 fps (embedded) + 4K (external)	1280 x 720 at 60 Hz, supports MIPI-DSI C-PHY and D-PHY
Camera	16 +16 IFEs, 32 MP, SHDR, MCTF, fisheye dewarp, EIS, MIPI-CSI 3 × 4 lanes, C-PHY, D-PHY	4 + 4 + 4 + 4/3 trios IFE 12 MP 30 fps ZSL
Video	4K60 (encode and decode) + 1080p	1080P60 decode + 1080P60 encode
Audio	V65B, 1190.4 MHz turbo, 512 KB L2, 16 KB L1 instruction cache, 16 KB L1 data cache	V73M, 16 K I cache/16K D cache/2 MB (L2/TCM) 1.34 GHz
Sensors	V65B, 1190.4 MHz turbo	V73M, 1.34 GHz
GPU	Qualcomm® Adreno™ GPU 615f	Adreno GPU 621
Storage/Peripherals	eMMC 5.1 USB 3.1 Type-C	eMMC 5.1 USB 3.1 Gen 1 Type-C eUSB 2.0 2x PCIe Gen 3 1-lane
Connectivity	WCN3980/WCN3990, 2 × 2/11ac/DB-S, Bluetooth 5.0	External 802.11be with WCN7851, 2 x 2, Bluetooth 5.3

Hardware Feature Comparison vs. SXR1130 (cont.)

Features	SXR1130	SAR1130P
Compute DSP	- Qualcomm® Hexagon™ V65 Qualcomm® Hexagon™ Vector eXtensions (HVX) 950 MHz Nominal (+ 1.2 GHz Turbo) 256 KB VTCM - Dual-HVX + 64 MMAC/MHz (16-bit fixed point) + 512 MMAC/MHz (8-bit fixed point) + 256 Mops/MHz (16-bit fixed point) - Scatter/Gather, DMA, UBWC	- Scalar processing V73 core with six hardware threads 2x HVX vector tightly coupled coprocessors (update CDSP slide) - Vector coprocessor optimized for pixel processing 1x matrix tightly coupled coprocessor - MAC arithmetic formats, 8-bit and 16-bit integer - Vector tightly coupled memory (VTCM) 2 MB - L2/TCM 1 MB
Audio DSP	- Hexagon v66K 864 MHz Nominal (1 GHz Turbo) 2 threads - L1 Instruction: 16 K L1 Data: 16 K L2: 512 K - Performance • 8 MMAC/MHz (16-bit fixed point) 16 MMAC/MHz (8-bit fixed point) • 4 MFLOP (SP)/MHz • 7.2 DMIPS/MHz • 14.4 Coremark/MHz	- Hexagon V66M processor can run up to 1.34 GHz in Turbo mode 2 cluster/4 hardware threads (2 threads per cluster) 16K I\$/16K D\$/2 MB (L2/TCM) - WCN785x (Bluetooth) - See <i>SAR1115P/SAR1130P Linux Android Audio Overview</i> (80-37610-50) for more details
Sensor DSP	- Hexagon V65B - 850 MHz nominal, 1.2 GHz Turbo - Two hardware threads - Shared with audio - Low Power Island - L1 16I/16D KB - Island memory – 1 MB	- Hexagon V73M – 1.34 GHz Turbo - Four hardware threads - Shared with audio - L1 16I/16D KB - Sensors supported: - Accelerometer/gyroscope/magnetometer - Proximity - Ambient light - PPG - Plus others per customer input
Machine intelligence	Compute DSP 1200 MHz, 512 kB L2 cache, 256 kB VTCM - Dual-HVX - Imaging and machine learning - Scatter/gather - Enhanced CNN support	Qualcomm® Hexagon™ Tensor Processor (HTP) with 2x Hexagon Vector eXtensions (HVX) and Hexagon Matrix eXtensions (HMX)
Security	Armv8-a hardware-based TrustZone (TZ) security environment	Trust management engine (TME) as the platform root of trust

SXR1130P Document Hierarchy

- Select the SAR1130P DEV product kit in Qualcomm-site to access the required documentation on SAR1130P
- See <https://www.qualcomm.com/products/mobile/snapdragon/xr-vr-ar/snapdragon-ar1-gen-1-platform#Documentation>

🏠 / Products / Mobile ▾ / XR, VR, AR, MR ▾ / Snapdragon AR1 Gen 1 Plat...

This product has not been publicly announced. Per your agreement, you have been granted access and all information is confidential.



Snapdragon AR1 Gen 1 Platform

Overview **Documentation** Software

All Documents ▾ 🔍 Search Product

602 results

RESULTS FOR:

Categories ^

☒ All Documents
☐ Hardware Documents
☐ Software Documents

FILTER BY

Software Distro & Tools ^
☐ Qualcomm® Package Manager

Classification ^
☐ Announcement

Data Sheet (13)

SAR1115P Data Sheet
Rev AE 80-36220-1 PDF

WCN7851 Tri-Band 2X2 MIMO DBS/HBS Wi-Fi 7 + Bluetooth 5.3 Data Sheet
Rev R 80-WL572-1 PDF

WCN7850 Tri-Band 2X2 MIMO DBS/HBS Wi-Fi 7 + Bluetooth 5.3 Data Sheet
Rev L 80-WL571-1 PDF

PMAR2130 Data Sheet
Rev AG 80-33949-1 PDF



Section 2

SAR1130P Reference Design Platforms

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

SAR1130P Reference Design

- See *AR1-Gen1 SAR1130P XR Development Platform (XDP) User Guide* (80-43766-135) for details.

Platform	Product Name	Connectivity
Development platform Key goal: software development	SAR1130P	WiFi6E 2x2 antennas

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Note: Contact the QTI sales team to order reference platforms.

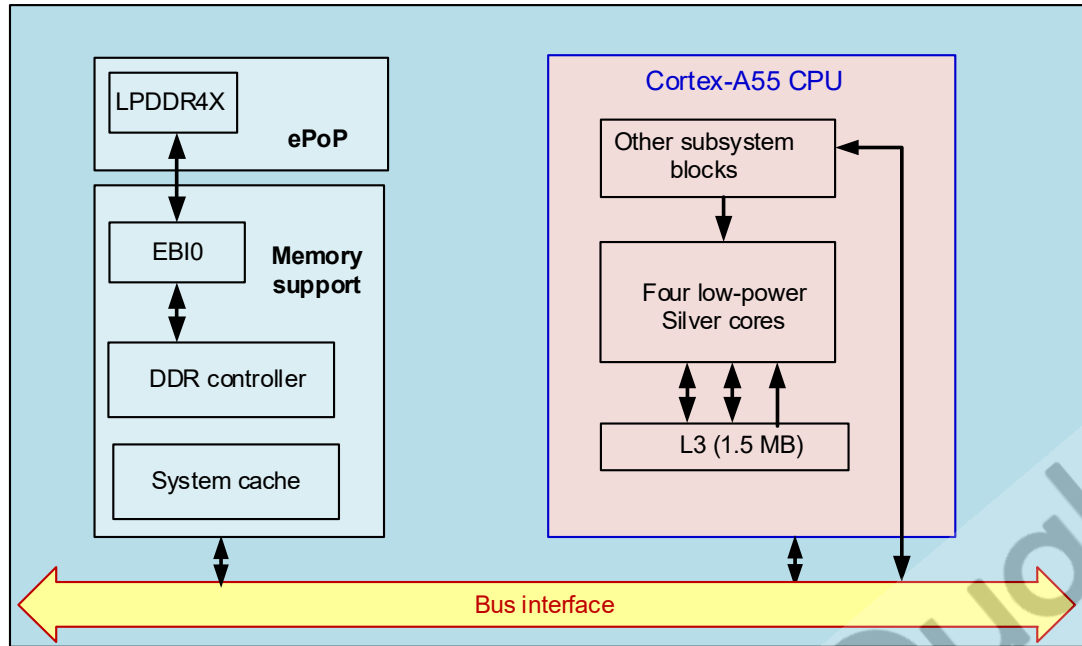


Section 3

SAR1130P Hardware Features – CPU

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

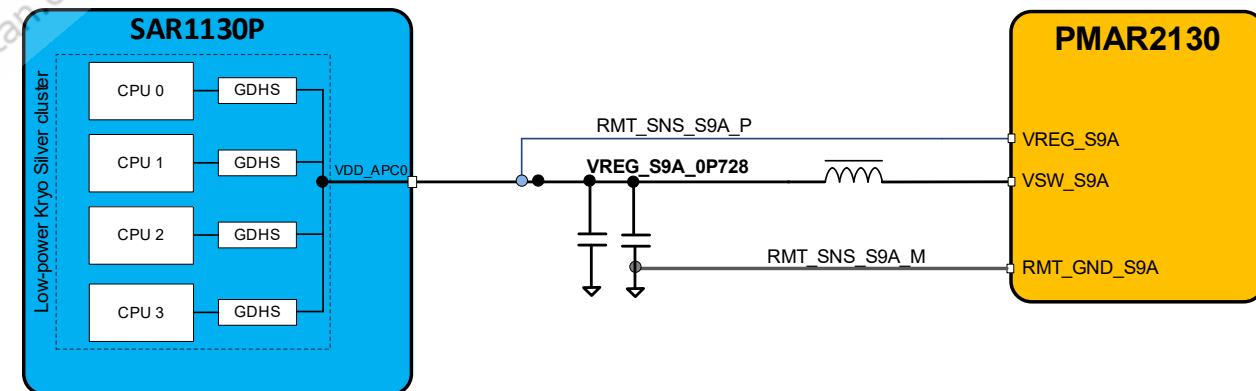
SAR1130P CPU Configuration



Features:

- Power-efficient Cortex-A55 CPU built on Arm v8 cortex technology
- Four Silver cores organized to optimize performance and power efficiency
- Dedicated L2 cache for each Silver core
- L3 shared cache of 1.5 MB

- PMAR2130 SMPS output S9A is routed as a separate trace to CPU Silver cluster
- Each CPU core inside the SAR1130P device has a global distributed head switch (GDHS) to power the core
- Place the 3-terminal (3T) output bulk capacitors as close as possible to the SAR1130P device to reduce series resistance and ensure transient performance



See *SAR1130P Digital Baseband Design and Layout Guidelines Training Slides* (80-43766-5) for details.



Section 4

SAR1130P Hardware Features – DSP

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Hexagon Tensor Processor

Fully software programmable

- Based on Hexagon Q6-DSP
- Leverage HVX (Vector Processing) ecosystem
- Architected around Vector and Matrix instruction primitives:
 - Vector processing and matrix processing are tightly coupled Q6-DSP coprocessors

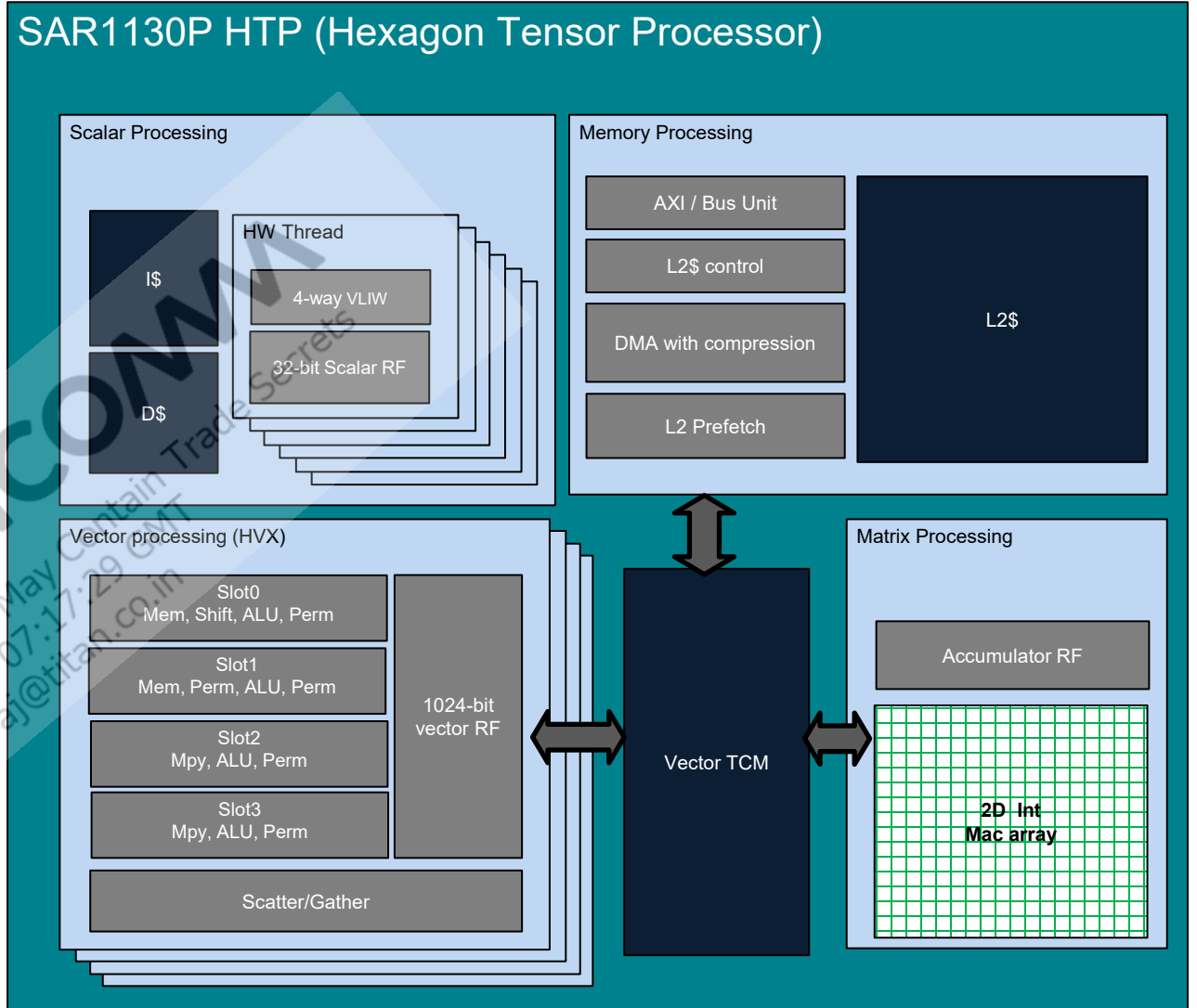
Hexagon Tensor Processor

- 6-Thread scalar DSP
- 2x HVX vector co-processors
 - Vector co-processor optimized for pixel processing
- 1x Integer HMX matrix co-processor
 - MAC arithmetic formats: 8-bit/16-bit integer

Memory subsystem

- 1 M L2 cache
- Large vector TCM
- User DMA
- Interlayer DDR compression for power bandwidth saving

SAR1130P HTP (Hexagon Tensor Processor)



See SAR1130P Digital Baseband Design and Layout Guidelines Training Slides (80-43766-5) for details.



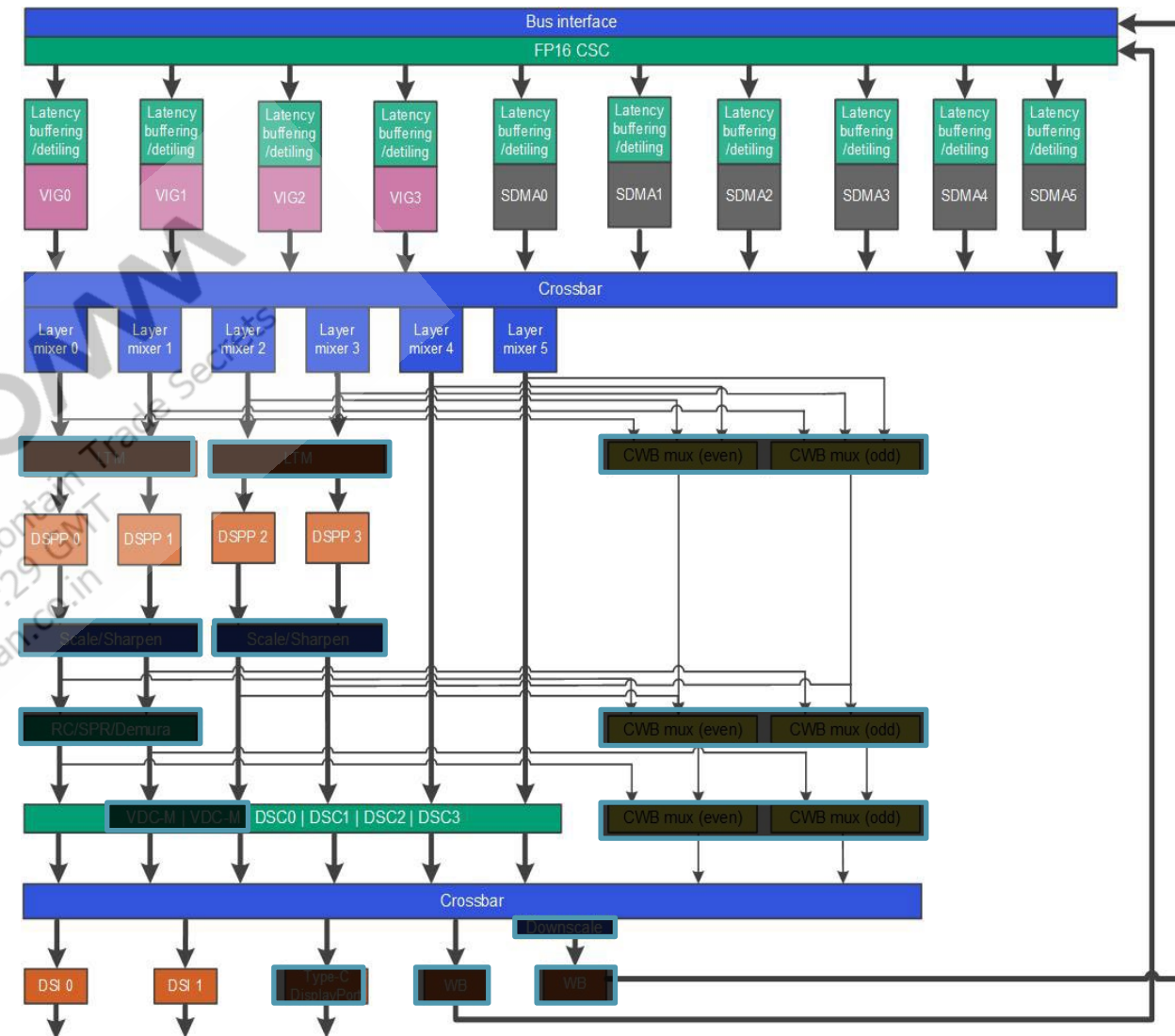
Section 5

SAR1130P Hardware Features – Display

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Adreno DPU1295 Overview

- Adreno Display processing unit (DPU)
 - Provides hardware-accelerated image processing, rapid transfer of image data to display interfaces, and enhanced on-screen image quality
- Source surface processor (VIG and Smart DMA source surface processor pipes (SSPP))
 - Performs color format conversion
- Layer mixer (LM)
 - Blends and mixes source surface together
- Destination surface processor (DSPP)
 - Conversion, correction, and adjustment based on panel characteristics
- VESA display compression (DSC)
 - Compressed display buffer being sent to display to reduce bandwidth and power consumption
- Display interface
 - Timing generator and interface connecting to display peripheral



See SAR1130P Linux Display Technical Overview (80-43035-83) for details.



Section 6

SAR1130P Hardware Features – GPU

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Adreno GPU 621 Features

- Features in GPU pipeline
 - A2D (dedicated 2D rasterizer) provides a low-power option for all the typical UI activities such as:
 - Common RGBA and YUV formats with different color depth
 - Blits
 - Scaling
 - Formats and tiling conversions
 - Color space conversions
 - The Adreno driver detects the type of UI operation and routes it to the shader processors or 2D rasterizer
- Architecture optimizations
 - For improved initialization time, Command processor (CP) micro code is not reloaded after power On or power Off of Adreno GPU
- Hardware clock gating
 - Gates unnecessary clocks based on the Adreno GPU activity
 - Improved block-specific clock gating
- Inter-frame power collapse for GX rail
- Adaptive clock distribution (ACD): It is a power-saving feature and supports the hardware block in the Adreno GPU. Its purpose is to mitigate the voltage droops on the graphics rail
- Hardware scheduler
 - Power management functions moved from kernel mode driver on APPS to GMU
 - Scheduler moved from KMD on APPS to GMU

See *SAR1130P Graphics Technical Overview* (80-43035-85) for details.

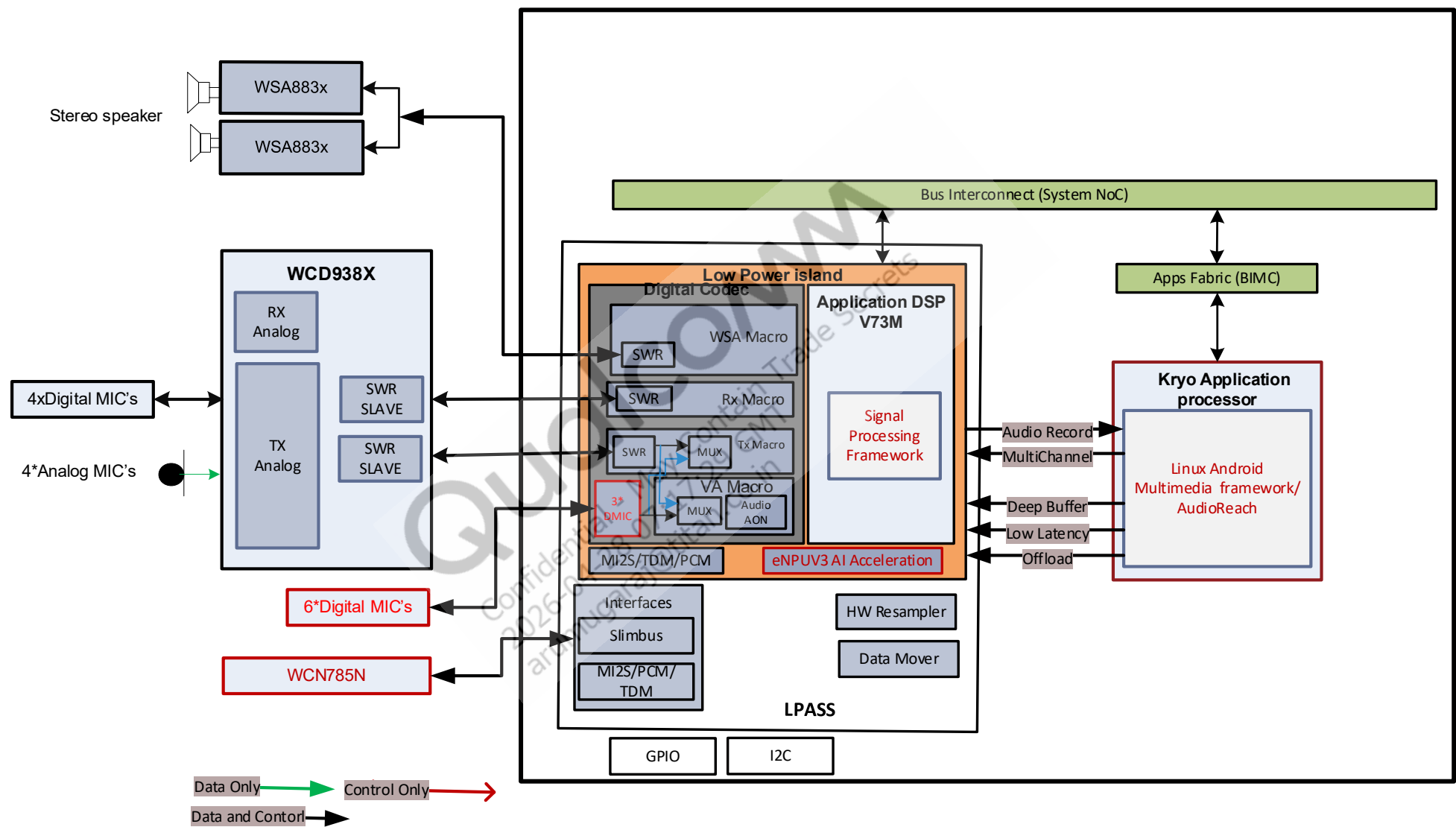


Section 7

SAR1130P Hardware Features – Audio

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

SAR1130P Audio System



See SAR1130P Audio Overview (80-37610-50) for details.

SAR1130P Audio System (cont.)

- SAR1115P/SAR1130P includes the next generation Qualcomm Snapdragon® Kryo785 application processors
- Audio system is comprised of hardware and software components:
 - Kryo application processor built on ARM Cortex technology.
 - Low power audio subsystem (LPASS), running audio/voice and voice activation (VA) algorithms, integrated with Dedicated Hexagon Processor (QDSP6) and low power island (LPI) with integrated digital audio codec
 - Hexagon V66M processor can run up to 1.34 GHz in Turbo mode
 - 2 cluster/4 hardware threads (2 threads per cluster)
 - 16K I\$/16K D\$/2 MB (L2/TCM)
 - Hardware accelerators to offload QDSP6
 - 3rd generation eNPU AI hardware accelerator for audio
 - Hardware resampler and hardware adaptive filter
 - One dedicated SLIMbus interface for WCN785x (Bluetooth)
 - Four MI2S/TDM interfaces to support third-party speaker
- WCN785x (Bluetooth)



Section 8

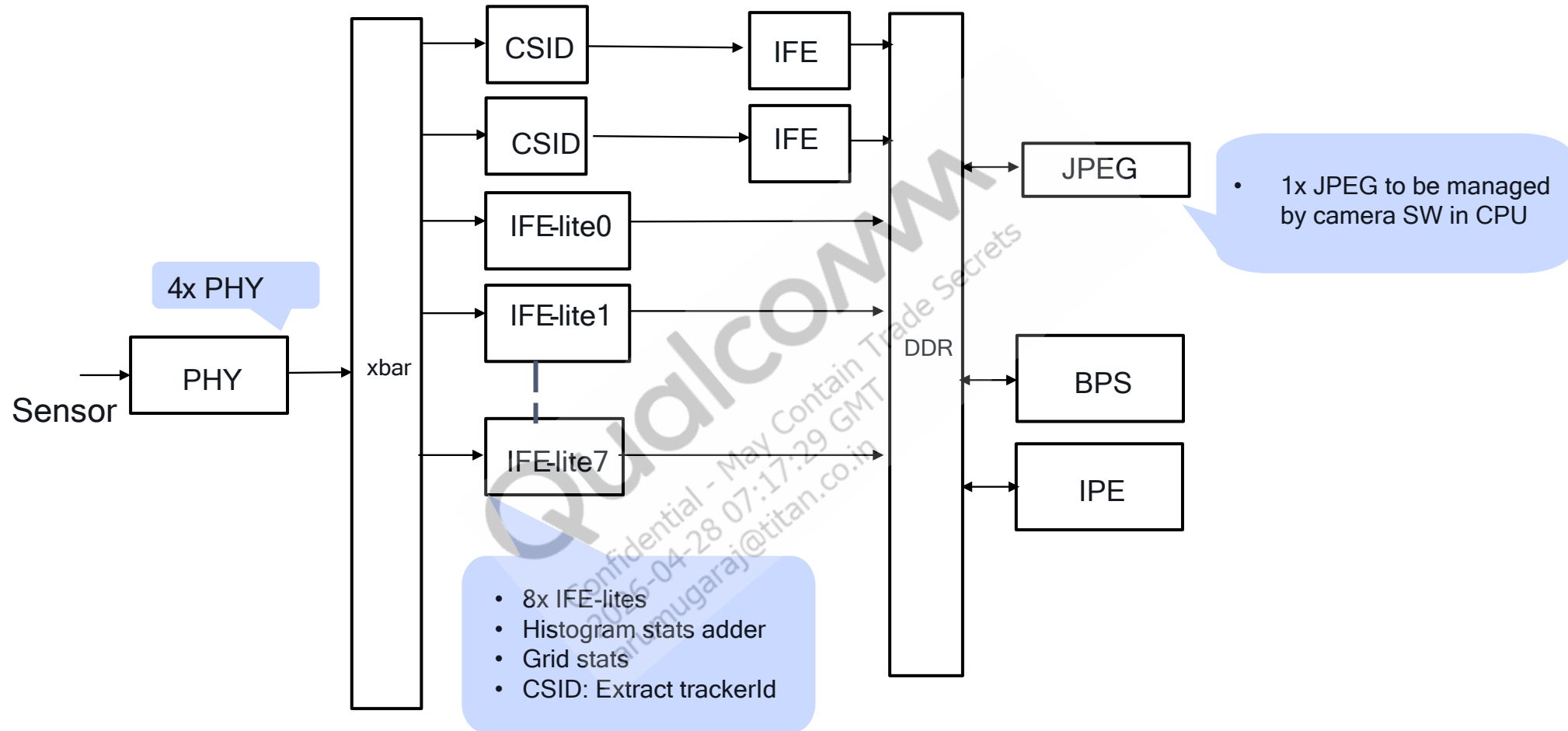
SAR1130P Hardware Features – Camera

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

SAR1130P Hardware Specifications

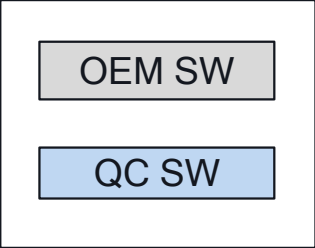
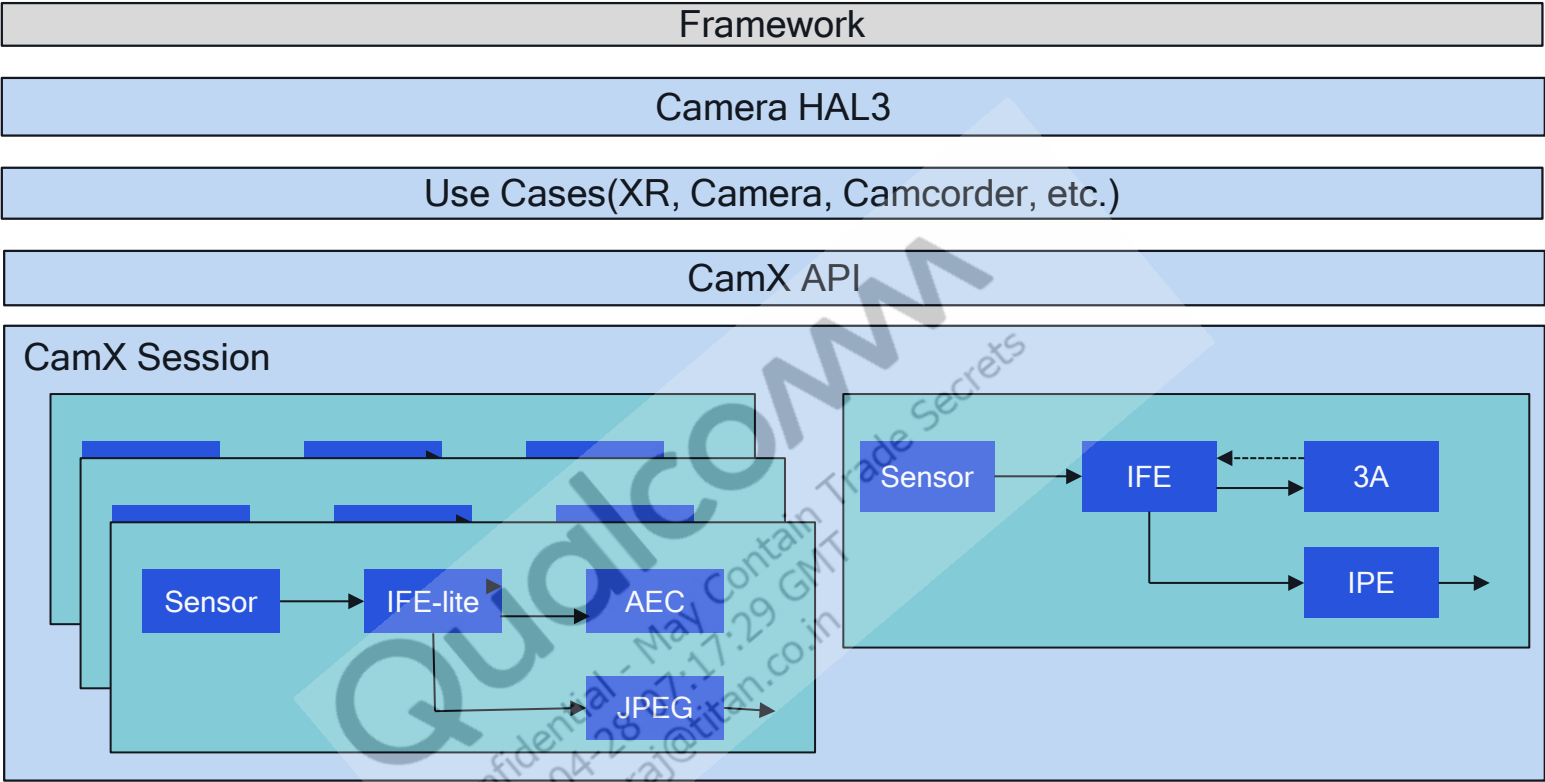
Features	SAR1130P
Maximum single camera resolution (hardware capability)	12 MP@ 30, ZSL (Bayer)
External Interface	DPHY 1.2/CPHY 1.2
Video recording (without EIS)	6.2 MP video encode with MCTF/FD
Video recording (with EIS)	720P video encode with MCTF/FD/EIS3.0
Live shot during video recording	3 MP
Real-time frontend engines	2x IFE + 8x IFE-lite
Backend processing engines	1x IPE, 1x BPS, 1x JPEG

Camera Hardware Architecture

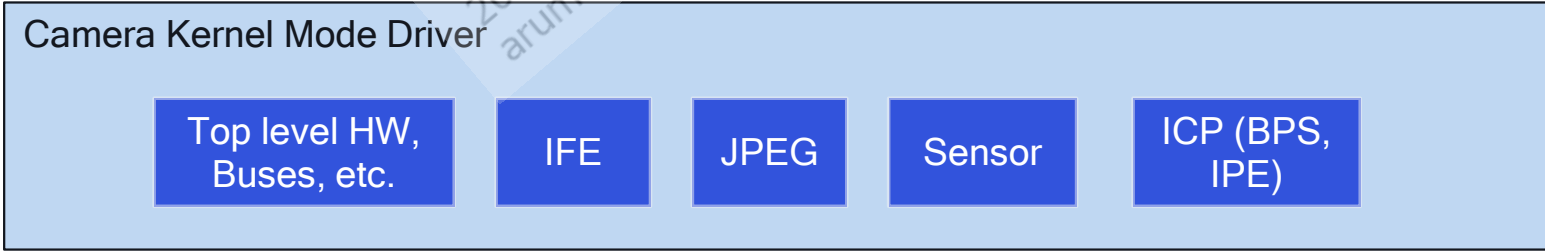


Camera Software Architecture

User Mode



Kernel



See SAR1130P Camera Overview (80-37610-82) for details.

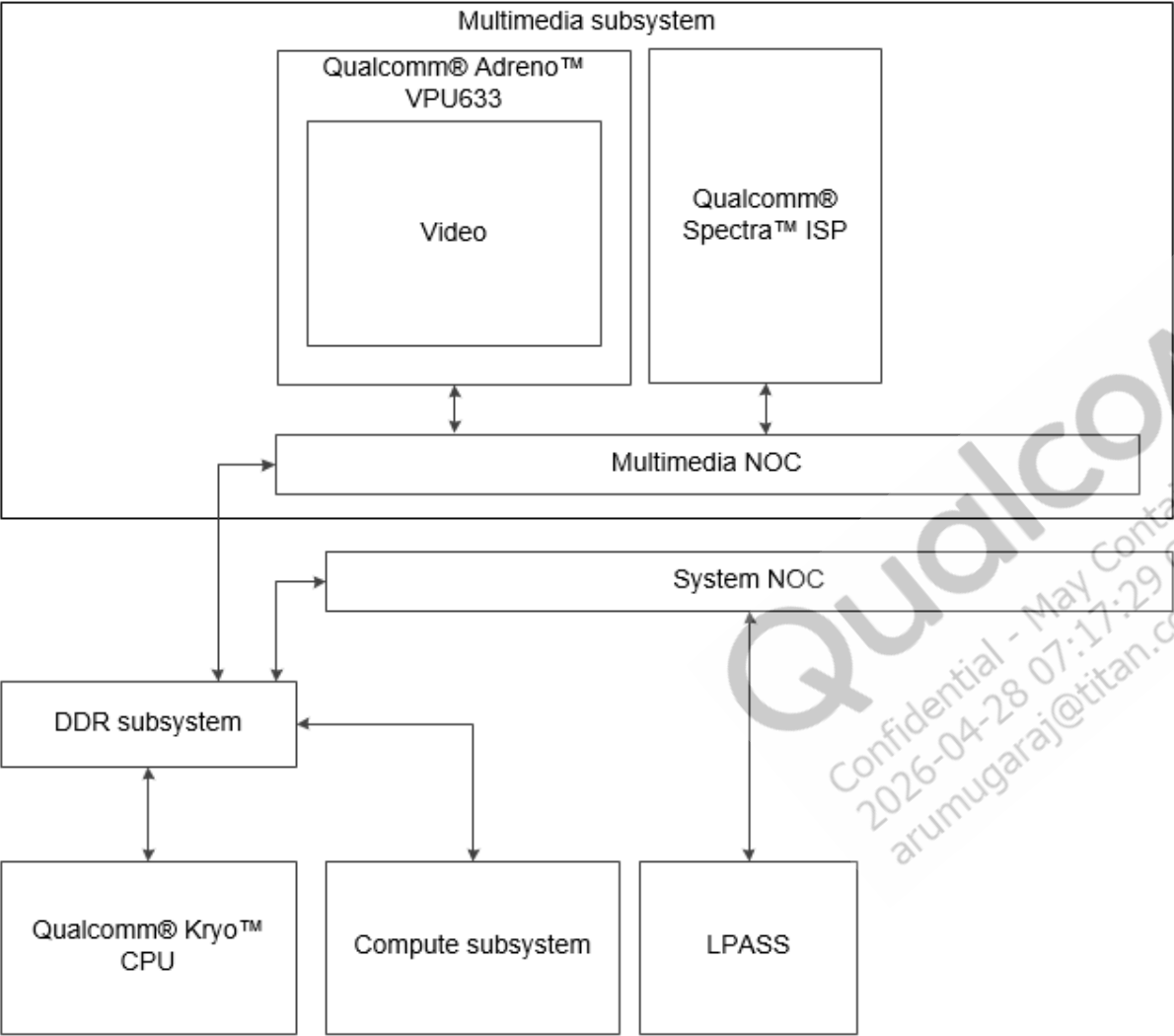


Section 9

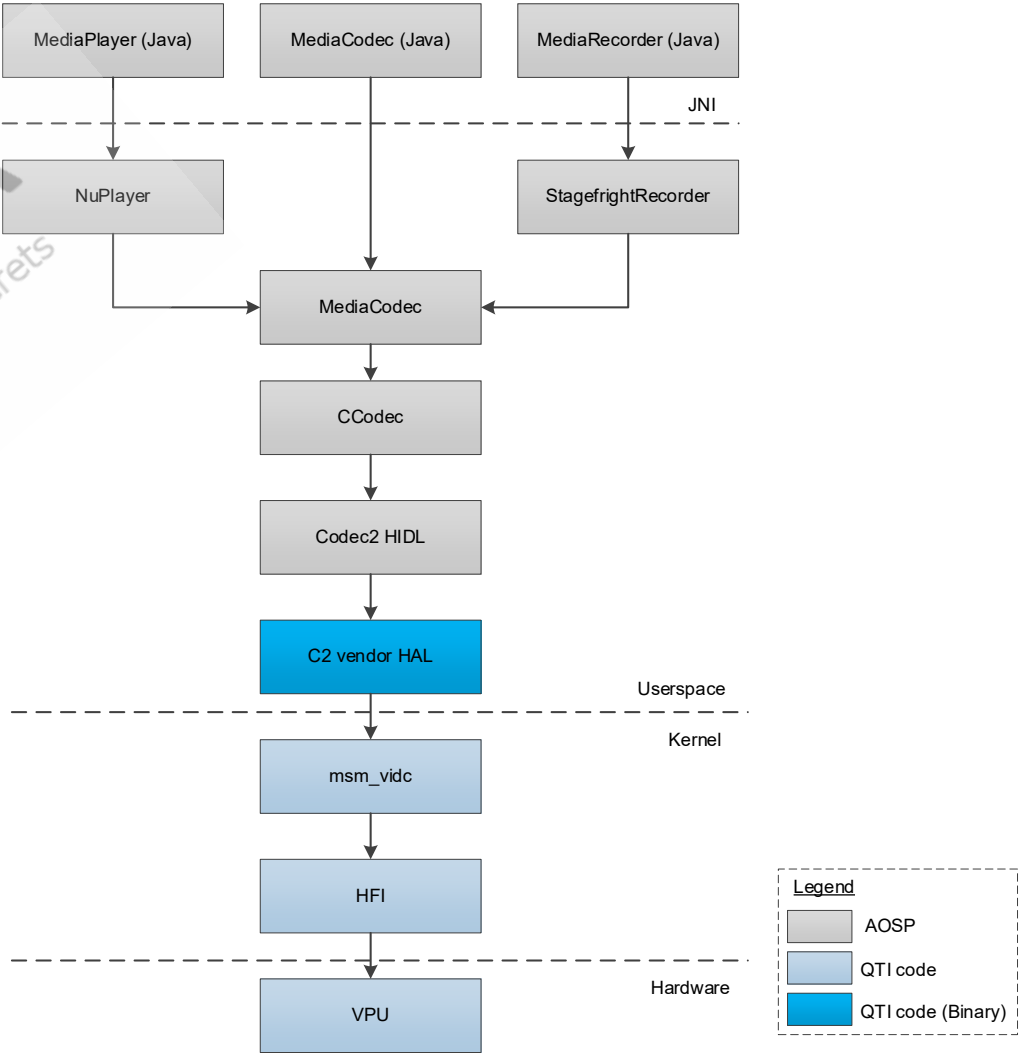
SAR1130P Hardware Features – Video

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

SAR1130P Video Architecture



Codec2 software stack



See SAR1130P Linux Android Video Technical Overview (80-37610-89) for details.



Section 10

SAR1130P Hardware Features – Security

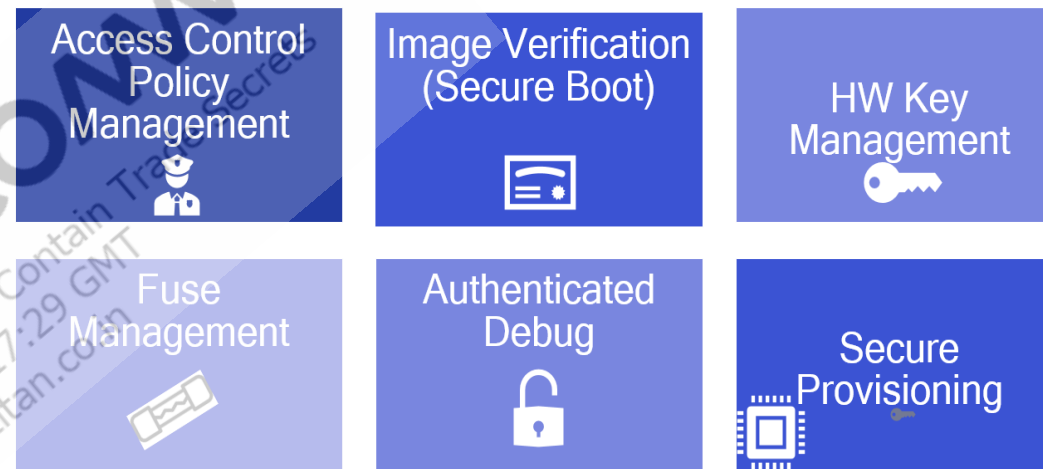
Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Trust Management Engine

TME has been designed as a subsystem providing the primary RoT for the SoC, offering security, and trust services to other subsystems of the SoC and to the parties that use it.

Services offered by TME to the SoC include:

- Image verification
- Hardware key management
- Secure provisioning
- Authenticated debug
- Access control management
- Fuse management



Trust management engine services

See *SAR1130P TrustZone and Security Technical Overview* (80-37610-40) for details.

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | May contain U.S. and international export controlled information | 80-43035-8 Rev. AA

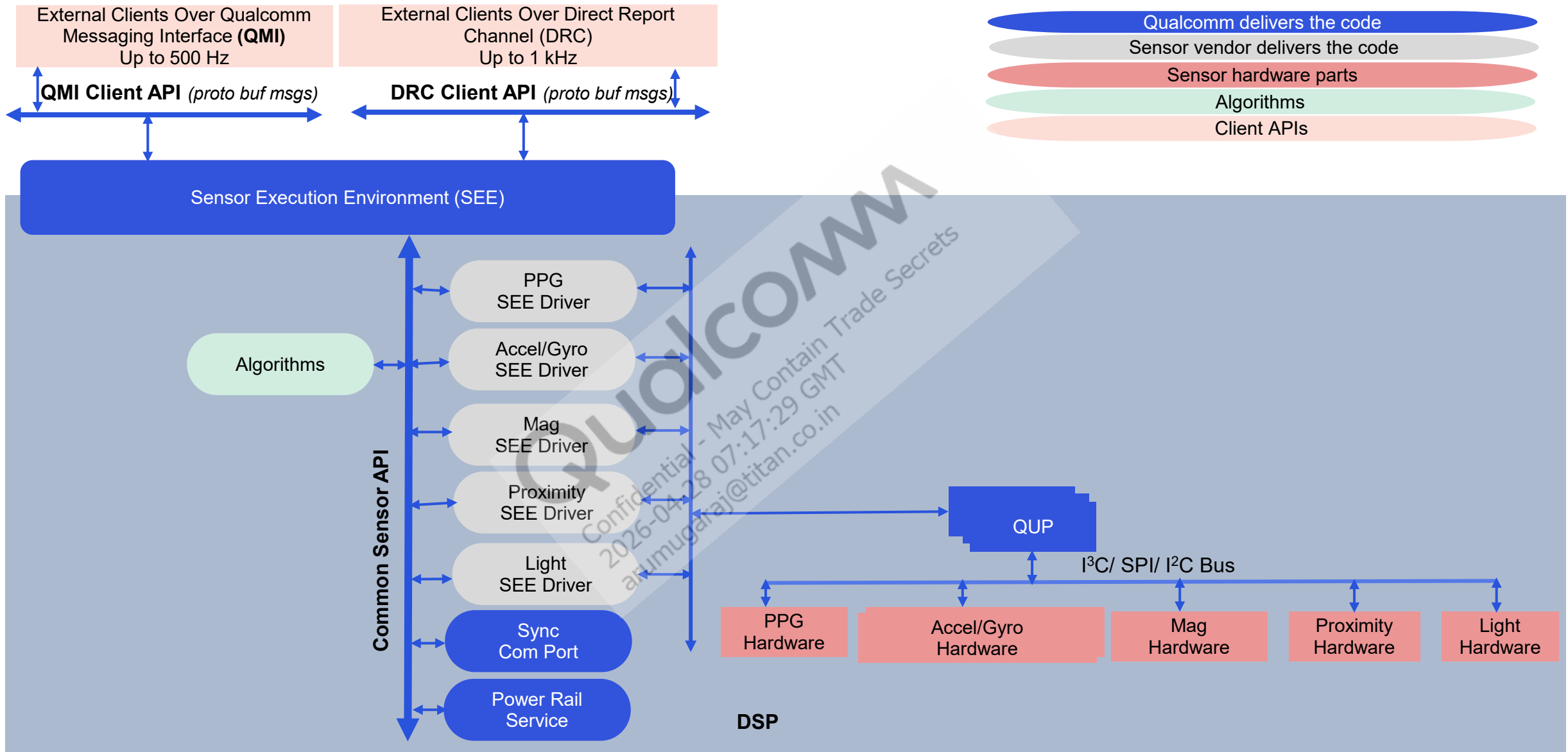


Section 11

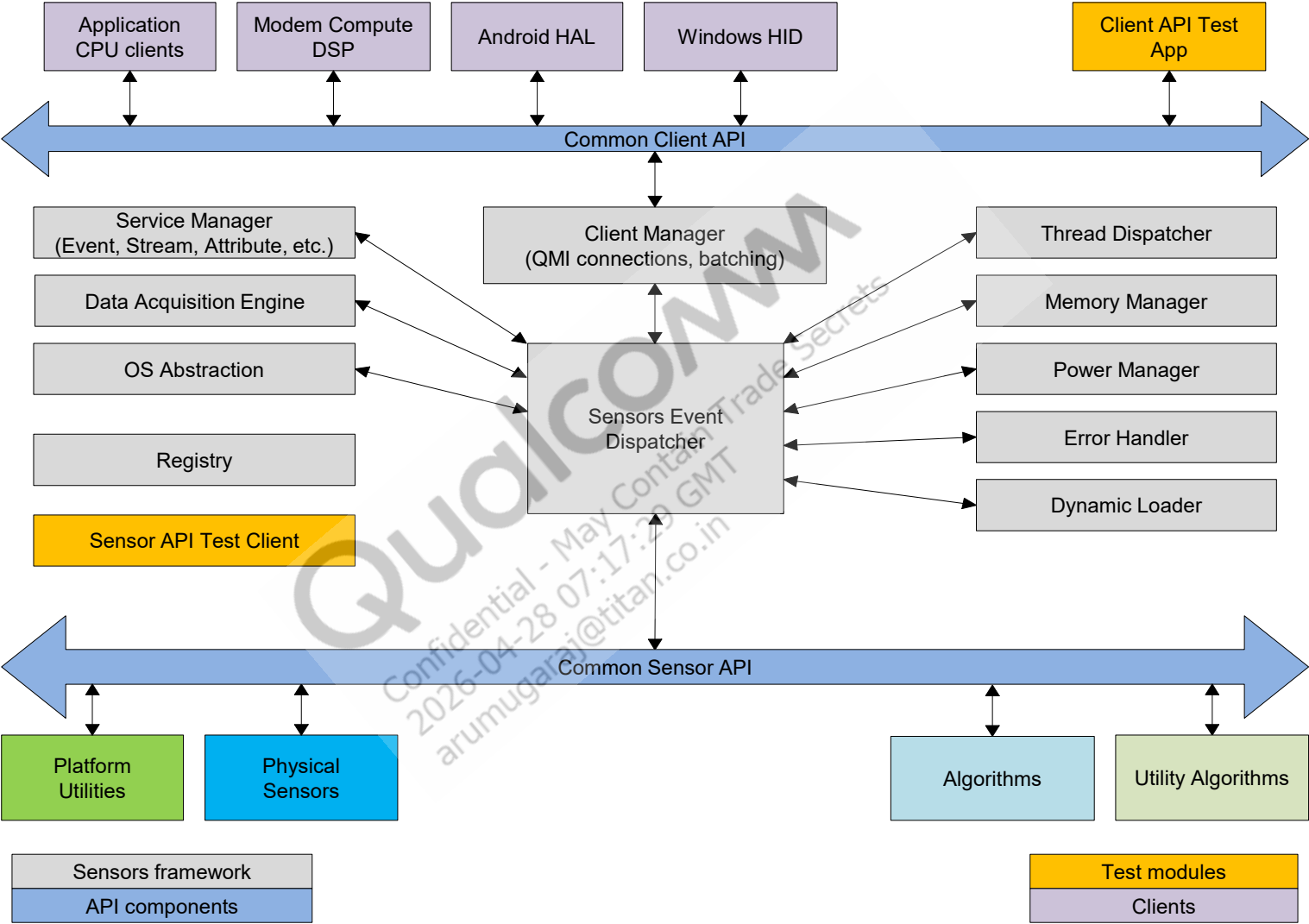
SAR1130P Sensor Modules

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

XR Modules – Sensors Architecture



SEE Block Diagram



See SAR1130P Sensors Overview (80-37610-24) for details.



Section 12

References

Qualcomm
Confidential - May Contain Trade Secrets
2026-04-28 07:17:29 GMT
arumugaraj@titan.co.in

Documents and Standards

Documents	
Title	Number
Qualcomm Technologies, Inc.	
<i>SAR1130P Digital Baseband Design and Layout Guidelines/Training Slides</i>	80-33594-5
<i>SAR1130P Data Sheet</i>	80-33594-1
<i>SAR1130P Graphics Technical Overview</i>	80-37611-85
<i>SAR1130P Linux Video Technical Overview</i>	80-37611-89
<i>SAR1130P Audio Overview</i>	80-37611-50
<i>SAR1130P Sensors Overview</i>	80-37611-24
<i>SAR1130P Camera Overview</i>	80-37610-82
<i>SAR1130P TrustZone and Security Technical Overview</i>	80-37610-40

Questions?

For additional information or to submit technical questions, go to: <https://www.qualcomm.com/support>

LEGAL INFORMATION

Your access to and use of this material, along with any documents, software, specifications, reference board files, drawings, diagnostics and other information contained herein (collectively this “Material”), is subject to your (including the corporation or other legal entity you represent, collectively “You” or “Your”) acceptance of the terms and conditions (“Terms of Use”) set forth below. If You do not agree to these Terms of Use, you may not use this Material and shall immediately destroy any copy thereof.

1) Legal Notice.

This Material is being made available to You solely for Your internal use with those products and service offerings of Qualcomm Technologies, Inc. (“**Qualcomm Technologies**”), its affiliates and/or licensors described in this Material, and shall not be used for any other purposes. If this Material is marked as “**Qualcomm Internal Use Only**”, no license is granted to You herein, and You must immediately (a) destroy or return this Material to Qualcomm Technologies, and (b) report Your receipt of this Material to qualcomm.support@qti.qualcomm.com. This Material may not be altered, edited, or modified in any way without Qualcomm Technologies’ prior written approval, nor may it be used for any machine learning or artificial intelligence development purpose which results, whether directly or indirectly, in the creation or development of an automated device, program, tool, algorithm, process, methodology, product and/or other output. Unauthorized use or disclosure of this Material or the information contained herein is strictly prohibited, and You agree to indemnify Qualcomm Technologies, its affiliates and licensors for any damages or losses suffered by Qualcomm Technologies, its affiliates and/or licensors for any such unauthorized uses or disclosures of this Material, in whole or part.

Qualcomm Technologies, its affiliates and/or licensors retain all rights and ownership in and to this Material. No license to any trademark, patent, copyright, mask work protection right or any other intellectual property right is either granted or implied by this Material or any information disclosed herein, including, but not limited to, any license to make, use, import or sell any product, service or technology offering embodying any of the information in this Material.

THIS MATERIAL IS BEING PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESSED, IMPLIED, STATUTORY OR OTHERWISE. TO THE MAXIMUM EXTENT PERMITTED BY LAW, QUALCOMM TECHNOLOGIES, ITS AFFILIATES AND/OR LICENSORS SPECIFICALLY DISCLAIM ALL WARRANTIES OF TITLE, MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, SATISFACTORY QUALITY, COMPLETENESS OR ACCURACY, AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. MOREOVER, NEITHER QUALCOMM TECHNOLOGIES, NOR ANY OF ITS AFFILIATES AND/OR LICENSORS, SHALL BE LIABLE TO YOU OR ANY THIRD PARTY FOR ANY EXPENSES, LOSSES, USE, OR ACTIONS HOWSOEVER INCURRED OR UNDERTAKEN BY YOU IN RELIANCE ON THIS MATERIAL.

Certain product kits, tools and other items referenced in this Material may require You to accept additional terms and conditions before accessing or using those items.

Technical data specified in this Material may be subject to U.S. and other applicable export control laws. Transmission contrary to U.S. and any other applicable law is strictly prohibited.

Nothing in this Material is an offer to sell any of the components or devices referenced herein.

This Material is subject to change without further notification.

In the event of a conflict between these Terms of Use and the *Website Terms of Use* on www.qualcomm.com, the *Qualcomm Privacy Policy* referenced on www.qualcomm.com, or other legal statements or notices found on prior pages of the Material, these Terms of Use will control. In the event of a conflict between these Terms of Use and any other agreement (written or click-through, including, without limitation any non-disclosure agreement) executed by You and Qualcomm Technologies or a Qualcomm Technologies affiliate and/or licensor with respect to Your access to and use of this Material, the other agreement will control.

These Terms of Use shall be governed by and construed and enforced in accordance with the laws of the State of California, excluding the U.N. Convention on International Sale of Goods, without regard to conflict of laws principles. Any dispute, claim or controversy arising out of or relating to these Terms of Use, or the breach or validity hereof, shall be adjudicated only by a court of competent jurisdiction in the county of San Diego, State of California, and You hereby consent to the personal jurisdiction of such courts for that purpose.

2) Trademark and Product Attribution Statements.

Qualcomm is a trademark or registered trademark of Qualcomm Incorporated. Arm is a registered trademark of Arm Limited (or its subsidiaries) in the U.S. and/or elsewhere. The Bluetooth® word mark is a registered trademark owned by Bluetooth SIG, Inc. Other product and brand names referenced in this Material may be trademarks or registered trademarks of their respective owners.

Snapdragon and Qualcomm branded products referenced in this Material are products of Qualcomm Technologies, Inc. and/or its subsidiaries. Qualcomm patented technologies are licensed by Qualcomm Incorporated.